CS 301: Database Management Systems & Architecture

An Exhaustive Rigorous Syllabus on Relational Theory, Storage Engines, Transaction Processing & Distributed Databases

Course Level:	Undergraduate (Core Computer Science)	**Term / Year:**	Academic Year 2026–2027
Prerequisites:	Data Structures & Algorithms, Systems Fundamentals	**Format:**	4 Credit Hours (3h Lecture + 3h Systems Lab / week)

1. COURSE PHILOSOPHY & ARCHITECTURAL FRAMEWORK

This course delivers a rigorous end-to-end exploration of database systems, spanning relational mathematical foundations, query optimization engineering, storage layer internals, transaction processing theory, and modern distributed storage architectures. The focus extends beyond high-level SQL querying to analyze how a database management system (DBMS) manages memory, disk I/O, concurrency control, crash recovery, and execution plan generation under constraints.

Students will build components of a relational database engine from scratch, gaining direct exposure to buffer pool management, B+ Tree indexing, dynamic cost-based query optimization, write-ahead logging (WAL), and distributed consensus protocols.

> **Core Pedagogical Principle:** *"Understand the data path from disk blocks to transaction execution."* Students evaluate systems trade-offs among ACID guarantees, isolation levels, serializability proofs, and hardware cache/disk constraints.

2. COMPREHENSIVE 16-WEEK CURRICULUM

WEEK	MODULE TOPIC	RIGOROUS CONTENT & THEORETICAL FOUNDATIONS
Week 1	**Database System Architecture & Data Models** CORE	DBMS architecture overview (Parser, Optimizer, Execution Engine, Buffer Pool, Storage Manager); ANSI-SPARC 3-schema architecture; Data Independence; Data models comparison (Relational, Document, Key-Value, Graph).
Week 2	**Relational Algebra & Tuple Relational Calculus** CORE	Formal Relational Model; Core Operators: Select (σ), Project (π), Rename (ρ), Union ($\cup$), Set Difference ($-$), Cartesian Product ($\times$); Extended Operators: Theta Join ($\bowtie_\theta$), Natural Join, Outer Joins, Aggregation; Tuple/ Domain Relational Calculus.
Week 3	**Declarative Querying: Advanced SQL Engineering** CORE	Complex SELECT statements, CTEs (Common Table Expressions & Recursive CTEs), Window functions (`OVER`, `PARTITION BY`, `RANK`), Subqueries, Correlated Subqueries, Set operations, Views, and Materialized Views.
Week 4	**Database Design & Entity-Relationship Modeling** CORE	Conceptual design using ER and Enhanced ER (EER) diagrams (Subclasses, Superclasses, Generalization, Specialization); Mapping ER/ EER schemes directly into relational schemas; Key constraints, Foreign keys, Referential integrity.

WEEK	MODULE TOPIC	RIGOROUS CONTENT & THEORETICAL FOUNDATIONS
Week 5	**Normalization & Functional Dependencies** ADVANCED	Formal Functional Dependencies (FDs), Armstrong's Axioms, Attribute Closure (X^+); Normal Forms: 1NF, 2NF, 3NF, Boyce-Codd Normal Form (BCNF); Lossless-Join decomposition, Dependency Preservation proofs, Multivalued Dependencies (4NF).
Week 6	**Storage Architecture & Disk Page Formats** CORE	Disk vs Memory hierarchies, Page layout models (Slotted-Page architecture), Record formats, Fixed vs Variable-length attributes, Columnar storage (DSM) vs Row storage (NSM) trade-offs, OS File I/O vs Direct I/O (`O_DIRECT`).
Week 7	**Midterm Examination & Engine Diagnostic Lab** EVAL	Comprehensive evaluation covering Relational Algebra, Advanced SQL, Normalization mathematical proofs, and Storage layout calculations.
Week 8	**Buffer Pool Management Architecture** ADVANCED	Buffer pool allocation, Frame replacement policies (LRU, Clock, LRU-K, 2Q), Dirty page flushing strategy, Pinning/Unpinning mechanics, Read-ahead prefetching, Memory-mapped files (`mmap`) vs Explicit Buffer Pools.
Week 9	**Indexing Algorithms: B+ Trees & Hash Indices** CORE	Tree Indexing: B+ Tree structure, operations (Search, Insertion, Node Split, Deletion, Node Merge), Prefix Compression, Clustered vs Non-Clustered Indices; Hash Indexing: Static Hashing, Extendible Hashing, Linear Hashing.
Week 10	**Query Processing & Physical Operator Execution** CORE	Iterator Execution Model (Volcano Model) vs Vectorized/Batch Processing; Physical Join Algorithms: Nested Loop Join, Block Nested Loop, Sort-Merge Join, Grace Hash Join; Sorting algorithms (External Merge Sort).
Week 11	**Query Optimization & Cost Estimation Models** ADVANCED	Relational Algebra equivalence rules, System R dynamic programming approach, Selinger optimizer model, Cost estimation formulas using histograms & statistics, Join order selection heuristics.
Week 12	**Transaction Theory & Concurrency Control** CORE	ACID properties; Atomicity, Consistency, Isolation, Durability; Conflict Serializability, Serialization Graphs; Two-Phase Locking (2PL), Strict 2PL, Rigorous 2PL; Deadlock Detection (Wait-For Graph) & Deadlock Prevention (Wait-Die, Wound-Wait).
Week 13	**Advanced Isolation Levels & Optimistic CC** ADVANCED	ANSI SQL Isolation Levels (Read Uncommitted, Read Committed, Repeatable Read, Serializable); Phenomena: Dirty Read, Non-Repeatable Read, Phantom Read, Write Skew; Multi-Version Concurrency Control (MVCC); Optimistic CC (OCC), Timestamp Ordering.
Week 14	**Crash Recovery Mechanics & Logging Protocols** ADVANCED	Write-Ahead Logging (WAL) protocol, Steal/No-Steal, Force/No-Force buffer management policies; ARIES Recovery Algorithm: Analysis Pass, Redo Pass, Undo Pass; Compensation Log Records (CLRs), Checkpointing techniques.
Week 15	**Distributed Databases, Consensus & NoSQL Systems** APPLIED	CAP Theorem, PACELC Theorem; Distributed Transactions: Two-Phase Commit (2PC) protocol; Distributed Consensus (Raft/Paxos overview); NoSQL Paradigms: Key-Value (Dynamo), Document (MongoDB), LSM-Tree Storage engines (RocksDB/Cassandra).
Week 16	**Capstone Defense & Database Systems Synthesis** APPLIED	Defense of custom-built Mini-DBMS (Implementing B+ Tree, Buffer Pool, and Query Execution Engine); Benchmarking workloads (TPC-C/TPC-H), final comprehensive written examination.

3. ASSESSMENT & GRADING FRAMEWORK

Evaluation assesses theoretical domain mastery, mathematical rigor in relational proofs, and hands-on database engine software engineering.

Component	Weight	Description & Objectives
Programming Assignments (Mini-DBMS)	30%	Iterative development of a relational storage engine component set: Buffer Pool Manager, B+ Tree Index, and Query Operator Execution Engine.
Theoretical Homework & Proofs	15%	Formal proofs of functional dependency closures, BCNF decompositions, conflict serializability graphs, and relational algebra rewrites.
Midterm Examination	20%	Comprehensive evaluation of Weeks 1–6 covering Relational Calculus, Normalization, SQL optimization, and Physical Page Layouts.
Database Capstone Defense	10%	Oral defense and benchmark performance analysis of the completed DBMS engine under concurrent workload constraints.
Final Comprehensive Exam	25%	Full-spectrum exam testing ARIES recovery algorithm execution, MVCC implementation details, query plan generation, and distributed transactions.

4. RECOMMENDED LITERATURE & REFERENCE MATERIALS

- **Silberschatz, A., Korth, H. F., & Sudarshan, S.** *Database System Concepts* (7th Ed.). McGraw-Hill.
- **Ramakrishnan, R., & Gehrke, J.** *Database Management Systems* (3rd Ed.). McGraw-Hill.
- **Hellerstein, J. M., & Stonebraker, M.** *Readings in Database Systems* (5th Ed. / "Red Book"). MIT Press.
- **Garcia-Molina, H., Ullman, J. D., & Widom, J.** *Database Systems: The Complete Book* (2nd Ed.). Pearson.